

Privacy-Preserving Integrity Proofs for Student-Society Voting: A Phase C Pilot of Project Simurgh at Macquarie University

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Abstract—Project Simurgh provides session-level integrity evidence for high-stakes digital interactions. This paper reports a Phase C consented shadow-mode pilot conducted alongside a real MQ Persian Society event preference poll. Thirty sessions completed the full consent-to-submit flow. In all 30 sessions, Simurgh collected no ballot choices, screen recordings, webcam or audio data, typed content, pasted content, or personal device identifiers. One additional consented session was withdrawn and excluded from analysis. After data collection closed, all write endpoints were enforced server-side with HTTP 410 Gone responses, demonstrating a reproducible collection-closure posture. The pilot demonstrates that privacy-preserving voting-session integrity evidence is achievable without observing or influencing the official vote. The study is voting-adjacent: it does not implement ballot cryptography, voter eligibility verification, coercion resistance, tally protection, or public-election certification.

Index Terms—privacy-preserving integrity, shadow-mode pilot, student-society voting, HMAC audit chain, consent-based telemetry, collection closure

I. INTRODUCTION

Student-society governance increasingly relies on digital voting tools. While these tools simplify logistics, they introduce questions about session integrity: can a participant verify that their session was conducted fairly without revealing who they voted for?

Project Simurgh is a research prototype designed to answer this question in the context of high-stakes digital interactions such as online examinations [1]. This paper reports a Phase C pilot in which Simurgh was deployed alongside a real MQ Persian Society event preference poll in shadow mode. Shadow mode means Simurgh ran in parallel with the official voting system, collected only session-level integrity metadata, and had no effect on the official election result.

The pilot had three goals:

- 1) Verify that the consent-to-submit flow completes without collecting ballot choices or privacy-sensitive data.

A voting-adjacent case study, not a production election-security claim. Research prototype only. No production deployment. Source: <https://github.com/Raouf128/Project-Simurgh>. Tag: v0.5.0-voting-pilot-phase-c-closeout.

- 2) Demonstrate that an HMAC audit chain correctly records session lifecycle events.
- 3) Show that server-side collection closure is enforceable and verifiable after the target dataset is reached.

Important scope note. This paper does not claim to be a study of election security, voting-system integrity, or electoral fraud prevention. It is a voting-adjacent case study of a privacy-preserving integrity prototype. The official election result was determined entirely by the society's own voting system.

II. RELATED WORK

A. End-to-End Verifiable Voting

Prior systems such as Helios [2] demonstrate web-based, open-audit voting protocols that use homomorphic encryption or mix-nets to allow voters to verify that their encrypted ballot was counted without revealing the plaintext choice. Civitas [3] and STAR-Vote [4] extend this line with coercion-resistance and risk-limiting-audit properties respectively.

This work differs in scope: Simurgh does not implement ballot encryption, voter-verifiable tallying, or public election certification. Instead, it evaluates whether *session-integrity* evidence can be collected around a voting-adjacent workflow without recording ballot choices or any surveillance data. Where end-to-end verifiable voting focuses on the integrity of the *outcome*, Simurgh focuses on the integrity of the *participation session*.

B. Remote and Internet Voting Security

Remote electronic voting introduces risks beyond ordinary web application security, including endpoint compromise, coercion, eligibility verification, ballot secrecy, and tally integrity. NIST guidance [5] identifies threat categories for remote electronic voting systems and recommends caution around internet-connected election infrastructure. Broader election-security consensus work [6] emphasises that vote collection, alteration, destruction, counting, and reporting are all distinct integrity concerns requiring separate mitigations.

Accordingly, this pilot is framed as a student-society research case study rather than a production online-voting system. It does not address eligibility, coercion resistance, malware-compromised endpoints, official tally integrity, or public-election certification.

C. Voting Standards, Privacy, and Auditability

The U.S. Election Assistance Commission adopted the Voluntary Voting System Guidelines (VVSG) 2.0 in 2021 [7], establishing updated requirements for privacy, accessibility, auditability, and software independence in voting systems. Modern voting-system guidance consistently emphasises the need for audit records that do not reveal individual ballot choices.

Simurgh aligns with the privacy-preserving and audit-evidence direction of this work — the HMAC audit chain records session lifecycle events without recording ballot content — but remains outside public-election certification scope and does not claim compliance with any voting-system standard.

D. Australian Technology-Assisted Voting Context

Technology-assisted voting has been explored in Australian jurisdictions, including iVote in New South Wales. The NSW Electoral Commission’s technology-assisted voting review [8] examined privacy, accessibility, and security trade-offs, recommending careful scoping and ongoing evaluation of voting technologies before broader deployment.

The Australian context remains cautious about technology-assisted voting, making small, bounded pilots useful for studying usability and privacy boundaries without affecting official outcomes. This pilot contributes a student-governance case study that tests consent, privacy boundaries, and session-integrity evidence in exactly that bounded scope.

E. Privacy-Preserving Telemetry and Data Minimisation

Simurgh’s ballot-field exclusion architecture is structurally related to data-minimisation principles and differential privacy [9]. The system collects the minimum metadata required to produce a session-integrity signal and structurally prevents ballot-choice data from reaching the server. This design echoes privacy-by-design principles [10] in which data that is never collected cannot be leaked.

F. Position of This Work

The contribution of this paper is not a new voting protocol. It is a bounded evaluation of privacy-preserving integrity evidence around a voting-adjacent workflow. The system remains blind to ballot choice and preserves a manual-review-only, non-disciplinary posture. It demonstrates that a shadow-mode consent-based pilot can produce verifiable session-integrity evidence at the student-society scale without any of the risks associated with internet voting in public elections.

III. SYSTEM DESIGN

A. Consent flow

Participants accessed a consent page that explained what Simurgh collects (session timestamps, focus-loss and paste counts, audit-chain validity signal, and an anonymous HMAC-hashed session code) and what it explicitly never collects (ballot choice, screen recordings, webcam/audio, microphone audio, typed text, clipboard content, raw process names, window titles, and device serial identifiers). Participation was voluntary; declining produced no consequence.

On consent, the server issued an anonymous participant code and a session token. No personally identifying information was collected at any step.

B. Ballot-field exclusion

The submit page discarded all radio button values in the browser before the `fetch` call, ensuring the ballot field was structurally absent from the request body. A second layer of protection was provided server-side: the `FORBIDDEN_BALLOT_FIELDS` set caused any submission body containing a known ballot-choice key to be rejected with `HTTP 400`, with no echo of the field value.

C. HMAC audit chain

Each session maintained an append-only HMAC audit chain recording lifecycle events: `STARTED`, `SUBMITTED`, and optionally `WITHDRAWN` or `REPORT_EXPORTED`. The chain was initialised on consent and verified on report export.

D. Collection-closure posture

After the target dataset was reached, the environment variable `SIMURGH_VOTING_PILOT_COLLECTION_CLOSED=true` was set. With this flag active:

- `POST /api/voting-pilot/consent/accept` returns 410 Gone.
- `POST /api/voting-pilot/submit` returns 410 Gone.
- `POST /api/voting-pilot/withdraw` returns 410 Gone.
- `GET /api/voting-pilot/:id/report` remains available, token-protected.

The closure is enforced before authentication middleware on write routes and cannot be bypassed with a valid session token.

IV. PHASE C PILOT

A. Study design

The pilot ran in shadow mode alongside the MQ Persian Society 2026 event preference poll. Participants were members of the society. Participation was voluntary and governed by the Phase C participant notice and data management addendum approved by the society executive on 2026-06-04.

B. Data management

Session data was held in an in-memory store. No data was persisted to disk. The store was cleared on server restart. Withdrawn sessions were excluded from report export. The withdrawn session (vp_4fcc741a) was confirmed blocked at the report endpoint with HTTP 403.

V. RESULTS

A. Dataset

TABLE I
SESSION COUNTS

Metric	Count
Consented pilot sessions	31
Submitted sessions (primary analysis)	30
Withdrawn sessions	1
Official vote impact	0
Ballot choices collected by Simurgh	0

B. Privacy assertions

TABLE II
PRIVACY ASSERTIONS — ALL 30 SUBMITTED SESSIONS

Assertion	Result
ballot_choice_recorded_by_simurgh	false
Ballot choice transmitted to server	No
Screen capture	No
Webcam / audio	No
Typed content	No
Pasted content	No
Personal device identifiers	No
Raw process names	No
Raw window titles	No
official_vote_impact	false
Privacy audit (tools/privacy-audit.mjs)	PASS

C. Integrity flow

All 30 submitted sessions completed the consent-to-submit flow. The HMAC audit chain recorded `STARTED` on consent and `SUBMITTED` on submit for each session. The withdrawn session's audit chain recorded `WITHDRAWN` and its report endpoint returned HTTP 403, confirming exclusion from analysis.

D. Safety gates

TABLE III
SAFETY GATES AT CLOSEOUT

Gate	Result
Node tests (npm test)	359/359 pass
npm audit (--audit-level=high)	0 high/critical
Privacy audit	PASS (52 files)
smoke-voting-pilot.sh	8/8 pass
security-audit-voting-pilot.sh	10/10 pass
smoke-voting-pilot-closed.sh	5/5 pass

E. Collection closure

After the 30-session target was reached, all write endpoints returned HTTP 410 Gone. The collection-closure smoke suite (5/5 gates pass) confirmed that consent/accept, submit, and withdraw are unreachable under the closed configuration, while report export remains available.

VI. DISCUSSION

A. What the results show

The pilot demonstrates that a consent-based shadow-mode integrity prototype can operate alongside a real student-society voting event without collecting ballot choices or any other privacy-sensitive information. The privacy assertions in Table II are enforced by system design — the ballot field is structurally absent from the server request body in every session — not solely by policy or participant compliance.

The server-side collection-closure mechanism provides a reproducible and auditable end-of-collection posture. Closure is verifiable by any party with network access to the server and is enforced independently of the client UI.

B. Limitations

- The 30-session dataset is a proof-of-concept pilot. No statistical power calculation was performed; generalisation of results to other contexts should be treated cautiously.
- Sessions were anonymous. No demographic or behavioural analysis of participants is possible or intended.
- The pilot ran in shadow mode only. No claim is made about the behaviour of Simurgh in a non-shadow deployment.
- The in-memory session store means session data did not survive server restarts. This is appropriate for a research prototype but is not a production data-management posture.

C. Non-claims

This pilot does not claim to:

- Secure or audit the official MQ Persian Society election result.
- Detect or prevent election fraud.
- Provide hardware-rooted attestation.
- Replace or improve the society's own voting system.
- Generalise to national, governmental, or binding elections.

VII. CONCLUSION

In a 30-session study pilot, Project Simurgh provided privacy-preserving voting-session integrity evidence without collecting ballot choices, screen recordings, webcam/audio, typed content, pasted content, or personal device identifiers. One additional consented session was withdrawn and excluded from analysis. After collection, all write endpoints were closed server-side with HTTP 410 responses.

The Phase C pilot establishes a clean research baseline (v0.5.0-voting-pilot-phase-c-closeout) for future work on privacy-preserving integrity proofs in voting-adjacent contexts.

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